

Laboratorium Metod Obliczeniowych w Nauce i Technice - ćwiczenie 4a

Ćwiczenie 4b - Aproksymacja Trygonometryczna

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AGH

Wydział Informatyki
Akademia Górniczo-Hutnicza

x_0	$\rho = 10^{-2}$	$\rho = 10^{-4}$	$\rho = 10^{-6}$	$\rho = 10^{-8}$	$\rho = 10^{-10}$	$\rho = 10^{-12}$	$\rho = 10^{-14}$	$\rho = 10^{-16}$	$\rho = 10^{-18}$
0.0	6.30e-03	2.13e-04	2.46e-07	3.94e-12	3.94e-12	6.31e-17	6.31e-17	6.31e-17	5.42e-18
0.1	4.92e-03	1.30e-04	9.13e-08	1.54e-12	1.54e-12	1.54e-12	2.13e-17	2.13e-17	5.42e-18
0.2	3.74e-03	7.60e-05	3.04e-08	5.23e-13	5.23e-13	5.23e-13	3.69e-18	3.69e-18	5.48e-18
0.3	2.32e-02	4.05e-05	8.33e-09	8.33e-09	1.44e-13	1.44e-13	2.93e-18	2.93e-18	2.93e-18
0.4	1.87e-02	1.75e-05	1.38e-09	1.38e-09	2.41e-14	2.41e-14	2.41e-14	4.99e-18	4.99e-18
0.5	1.25e-02	3.66e-06	1.01e-11	1.01e-11	1.01e-11	1.80e-16	1.80e-16	1.80e-16	5.42e-18
0.6	1.74e-03	1.65e-05	1.20e-09	1.20e-09	2.09e-14	2.09e-14	2.09e-14	5.04e-18	5.04e-18
0.7	1.18e-02	2.93e-06	3.67e-12	3.67e-12	3.67e-12	5.83e-17	5.83e-17	5.83e-17	5.42e-18
0.8	1.93e-02	1.99e-05	1.84e-09	1.84e-09	3.19e-14	3.19e-14	4.88e-18	4.88e-18	4.88e-18
0.9	1.06e-02	1.91e-06	1.91e-06	1.31e-11	1.31e-11	2.23e-16	2.23e-16	2.23e-16	5.42e-18
1.0	1.35e-02	5.01e-06	5.09e-11	5.09e-11	5.09e-11	8.90e-16	8.90e-16	5.42e-18	5.42e-18
1.1	4.88e-03	1.29e-04	8.86e-08	1.50e-12	1.50e-12	1.50e-12	2.05e-17	2.05e-17	5.42e-18
1.2	1.76e-02	1.38e-05	8.11e-10	8.11e-10	1.41e-14	1.41e-14	1.41e-14	5.20e-18	5.20e-18
1.3	1.78e-02	1.45e-05	9.09e-10	9.09e-10	1.58e-14	1.58e-14	1.58e-14	5.15e-18	5.15e-18
1.4	3.67e-03	7.29e-05	2.80e-08	4.82e-13	4.82e-13	4.82e-13	2.98e-18	2.98e-18	5.48e-18
1.5	2.10e-02	2.75e-05	3.68e-09	3.68e-09	6.38e-14	6.38e-14	4.34e-18	4.34e-18	4.34e-18
1.6	2.06e-02	2.54e-05	3.10e-09	3.10e-09	5.39e-14	5.39e-14	4.50e-18	4.50e-18	4.50e-18
1.7	nan	nan	nan	nan	nan	nan	nan	nan	nan
1.8	nan	nan	nan	nan	nan	nan	nan	nan	nan
1.9	nan	nan	nan	nan	nan	nan	nan	nan	nan
2.0	3.10e-03	5.22e-05	1.41e-08	1.41e-08	2.43e-13	2.43e-13	1.19e-18	1.19e-18	5.42e-18

Tabela 1: wyniki dla stopu fx

x_0	$\rho = 10^{-2}$	$\rho = 10^{-4}$	$\rho = 10^{-6}$	$\rho = 10^{-8}$	$\rho = 10^{-10}$	$\rho = 10^{-12}$	$\rho = 10^{-14}$	$\rho = 10^{-16}$	$\rho = 10^{-18}$
0.0	6.30e-03	2.13e-04	2.46e-07	3.94e-12	3.94e-12	6.31e-17	6.31e-17	6.31e-17	5.42e-18
0.1	4.92e-03	1.30e-04	9.13e-08	1.54e-12	1.54e-12	1.54e-12	2.13e-17	2.13e-17	5.42e-18
0.2	3.74e-03	7.60e-05	3.04e-08	5.23e-13	5.23e-13	5.23e-13	3.69e-18	3.69e-18	5.48e-18
0.3	2.32e-02	4.05e-05	8.33e-09	8.33e-09	1.44e-13	1.44e-13	2.93e-18	2.93e-18	2.93e-18
0.4	1.87e-02	1.75e-05	1.38e-09	1.38e-09	2.41e-14	2.41e-14	2.41e-14	4.99e-18	4.99e-18
0.5	1.25e-02	3.66e-06	1.01e-11	1.01e-11	1.01e-11	1.80e-16	1.80e-16	1.80e-16	5.42e-18
0.6	1.74e-03	1.65e-05	1.20e-09	1.20e-09	2.09e-14	2.09e-14	2.09e-14	5.04e-18	5.04e-18
0.7	1.18e-02	2.93e-06	3.67e-12	3.67e-12	3.67e-12	5.83e-17	5.83e-17	5.83e-17	5.42e-18
0.8	1.93e-02	1.99e-05	1.84e-09	1.84e-09	3.19e-14	3.19e-14	4.88e-18	4.88e-18	4.88e-18
0.9	1.06e-02	1.91e-06	1.91e-06	1.31e-11	1.31e-11	2.23e-16	2.23e-16	2.23e-16	5.42e-18
1.0	1.35e-02	5.01e-06	5.09e-11	5.09e-11	5.09e-11	8.90e-16	8.90e-16	5.42e-18	5.42e-18
1.1	4.88e-03	1.29e-04	8.86e-08	1.50e-12	1.50e-12	1.50e-12	2.05e-17	2.05e-17	5.42e-18
1.2	1.76e-02	1.38e-05	8.11e-10	8.11e-10	1.41e-14	1.41e-14	1.41e-14	5.20e-18	5.20e-18
1.3	1.78e-02	1.45e-05	9.09e-10	9.09e-10	1.58e-14	1.58e-14	1.58e-14	5.15e-18	5.15e-18
1.4	3.67e-03	7.29e-05	2.80e-08	4.82e-13	4.82e-13	4.82e-13	2.98e-18	2.98e-18	5.48e-18
1.5	2.10e-02	2.75e-05	3.68e-09	3.68e-09	6.38e-14	6.38e-14	4.34e-18	4.34e-18	4.34e-18
1.6	2.06e-02	2.54e-05	3.10e-09	3.10e-09	5.39e-14	5.39e-14	4.50e-18	4.50e-18	4.50e-18
1.7	nan	nan	nan	nan	nan	nan	nan	nan	nan
1.8	nan	nan	nan	nan	nan	nan	nan	nan	nan
1.9	nan	nan	nan	nan	nan	nan	nan	nan	nan
2.0	3.10e-03	5.22e-05	1.41e-08	1.41e-08	2.43e-13	2.43e-13	1.19e-18	1.19e-18	5.42e-18

Tabela 2: wyniki dla stopu fx